تربية وتعليم

مدرسة الخالدية الحلقة الثالثة بنين الشرقية

AL KHALIDIA BOYS' SCHOOL · C3 EASTERN

الميثاق الوطني للتعليم

END OF TERM 1 · ACADEMIC YEAR 2026 – 2027

Grade 11 Advanced

Exam Coverage Compilation

تجميعة هيكل امتحان نهاية الفصل الدراسي الأول

Reveal Integrated Math III | 25 questions | MCQ 1–20 · FRQ 21–25 | 150 min · /100

Draft coverage — for teachers' feedback only · not for students

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بالميثاق نبلغ الآفاق

Grade 11 Advanced · Coverage at a glance

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15	**Lesson 9-1** Angles and Angle Measure	Draw and find angles in standard position	Example 3	**P418**
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Grade 11 Advanced · Coverage at a glance

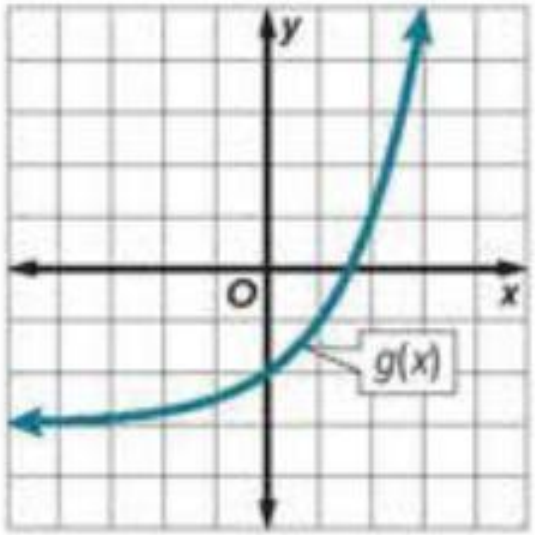
Q	LESSON	LEARNING OUTCOME	EXAMPLE / EXERCISE	PAGE
19	**Lesson 9-5** Graphing Other Trigonometric Functions	Describe and graph trigonometric functions other than sine, cosine, and tangent	Exercises 7-15	**P461**
20	**Lesson 9-7** Inverse Trigonometric Functions	Find values of angle measures by using inverse trigonometric functions	Exercises 1-9	**P475**
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23	**Lesson 7-4** Graphing Rational Functions	Graph rational functions with oblique asymptotes and point discontinuity	Example 4	**P341**
24	**Lesson 8-4** Normal Distributions	(a) Find the area under normal distribution curves. (b) Find probabilities for normal distributions and find data values given probabilities. (The table will be provided in the question)	Example 5 · Exercise 14	**P397 · P402**
25	**Lesson 9-1** Angles and Angle Measure	(a) Convert between degrees and radian measures and find arc lengths by using central angles. (b) Graph horizontal translations of trigonometric functions. (No need for a graphing calculator)	Example 6 · Exercises 52, 53 · Exercises 7-12	**P420 · P422 · P469**

◆ LESSON 5-1 · Graphing Exponential Functions

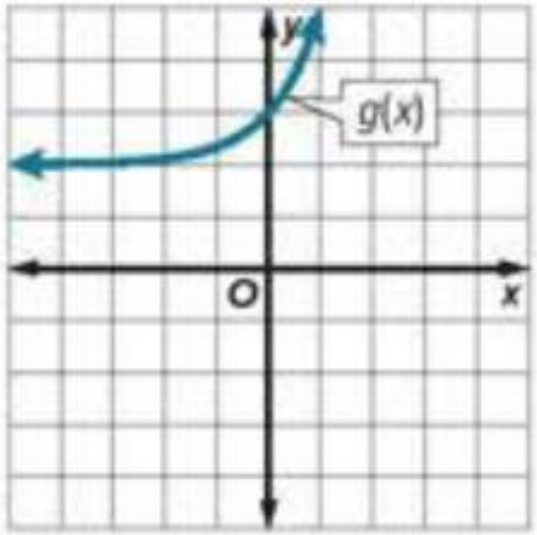
Example 3

Identify the value of *k* and write a function *g*(*x*) for each graph as it relates to *f*(*x*).

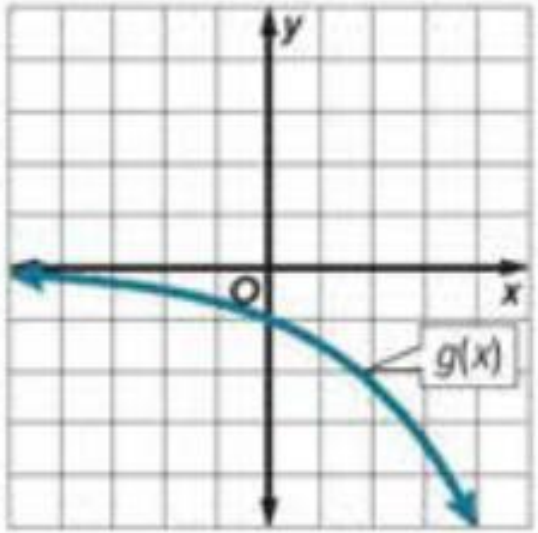
11. $f(x) = 2^x; g(x) = f(x) + k$



12. $f(x) = 3^x; g(x) = f(x) + k$



13. $f(x) = \left(\frac{3}{2}\right)^x; g(x) = k \cdot f(x)$



◆ LESSON 5-2 · Solving Exponential Equations and Inequalities

Mixed Exercises

Solve each equation or inequality.

23. $625 \geq 5^{a+8}$

29. $\left(\frac{1}{64}\right)^{c-2} < 32^{2c}$

32. $\left(\frac{1}{9}\right)^{3t+5} \leq \left(\frac{1}{243}\right)^{t-6}$

34. $10^{5b+2} > 1000$

Example 1

Write each equation in exponential form.

1. $\log_{15} 225 = 2$

2. $\log_3 \frac{1}{27} = -3$

3. $\log_5 \frac{1}{25} = -2$

4. $\log_3 243 = 5$

5. $\log_4 64 = 3$

6. $\log_4 32 = \frac{5}{2}$

Example 2

Write each equation in logarithmic form.

7. $2^7 = 128$

8. $3^{-4} = \frac{1}{81}$

9. $7^{-2} = \frac{1}{49}$

10. $\left(\frac{1}{7}\right)^3 = \frac{1}{343}$

11. $2^9 = 512$

12. $64^{\frac{2}{3}} = 16$

◆ LESSON 6-2 · Properties of Logarithms

Example 2

Solve each equation.

7. $\log_4 (2x^2 - 4) = \log_4 2x$

8. $\log_5 (x^2 - 6) = \log_5 x$

9. $\log_3 (x^2 - 8) = \log_3 2x$

10. $\log_4 (2x^2 - 20) = \log_4 6x$

11. $\log_2 (6x^2 + 1) = \log_2 5x$

12. $\log_6 (6x^2 - 3) = \log_6 7x$

Example 5 Change of Base Formula

Evaluate $\log_2 11$ by writing it in terms of common logarithms. Round to the nearest ten-thousandth.

$$\log_2 11 = \frac{\log_{10} 11}{\log_{10} 2}$$ Change of Base Formula

$$\approx 3.4594$$ Use a calculator.

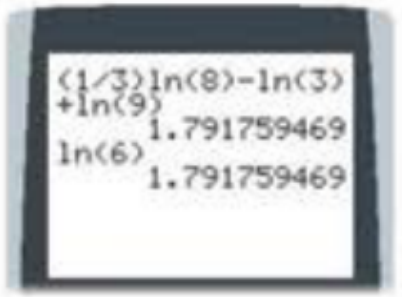
Example 3 Simplify Logarithmic Expressions

Write $\frac{1}{3}\ln 8 - \ln 3 + \ln 9$ as a single logarithm.

$\frac{1}{3}\ln 8 - \ln 3 + \ln 9$	Original expression
$= \ln 8^{\frac{1}{3}} - \ln 3 + \ln 9$	Power Property of Logarithms
$= \ln 2 - \ln 3 + \ln 9$	$8^{\frac{1}{3}} = \sqrt[3]{8}$ or 2
$= \ln \frac{2}{3} + \ln 9$	Quotient Property of Logarithms
$= \ln 6$	Product Property of Logarithms

Check

Use a calculator to verify the solution.



Because both entries result in the same value, $\frac{1}{3}\ln 8 - \ln 3 + \ln 9 = \ln 6$.

Check

Write $\ln 7 - \frac{1}{3}\ln 27 + \ln 6$ as a single logarithm.

◆ LESSON 7-1 · Multiplying and Dividing Rational Expressions

Example 1 Simplify a Rational Expression

Simplify $\frac{x^2 - 2x - 24}{2x^3 + 6x^2 - 8x}$, and state when the original expression is undefined.

Part A Simplify.

$$\frac{x^2 - 2x - 24}{2x^3 + 6x^2 - 8x} = \frac{x^2 - 2x - 24}{2x(x^2 + 3x - 4)}$$ Distributive Property

$$= \frac{(x + 4)(x - 6)}{2x(x + 4)(x - 1)}$$ Factor the numerator and denominator.

$$= \frac{(x - 6)\ \cancel{(x + 4)}^{1}}{2x(x - 1)\ \cancel{(x + 4)}_{1}}$$ Eliminate common factors.

$$= \frac{x - 6}{2x(x - 1)}$$ Simplify.

Part B State when the expression is undefined.

The expression is undefined when the denominator is equal to 0.

Because the question asks when the original expression is undefined, consider the denominator before the common factors are eliminated. Thus, the expression is undefined when $2x(x + 4)(x - 1) = 0$. By the Zero Product Property, the expression is undefined when $2x = 0$ or $x = 0$, when $x + 4 = 0$ or $x = -4$, and when $x - 1 = 0$ or $x = 1$.

◆ LESSON 7-2 · Adding and Subtracting Rational Expressions

Examples 4 and 5

Simplify each expression.

16. $\dfrac{\frac{2}{x-3}+\frac{3x}{x^2-9}}{\frac{3}{x+3}-\frac{4x}{x^2-9}}$

17. $\dfrac{\frac{4}{x+5}+\frac{9}{x-6}}{\frac{5}{x-6}-\frac{8}{x+5}}$

18. $\dfrac{\frac{5}{x+6}-\frac{2x}{2x-1}}{\frac{x}{2x-1}+\frac{4}{x+6}}$

19. $\dfrac{\frac{8}{x-9}-\frac{x}{3x+2}}{\frac{3}{3x+2}+\frac{4x}{x-9}}$

◆ LESSON 7-3 · Graphing Reciprocal Functions

Example 4

Graph each function. State the domain and range.

13. $f(x) = \frac{1}{x+3} - 3$

14. $f(x) = \frac{-1}{x+5} - 6$

15. $f(x) = \frac{-1}{x+1} + 3$

16. $f(x) = \frac{1}{x+4} - 2$

◆ LESSON 7-6 · Solving Rational Equations and Inequalities

Example 2 Solve a Rational Equation with an Extraneous Solution

Solve $\frac{2m}{m-4} - \frac{m^2+7m+4}{3m^2-18m+24} = \frac{4m}{3m-6}$.

The LCD for the terms is $(m-4)(3m-6)$.

$$\frac{2m}{m-4} - \frac{m^2+7m+4}{3m^2-18m+24} = \frac{4m}{3m-6}$$

$$\frac{(m-4)(3m-6)(2m)}{(m-4)} - \frac{(m-4)(3m-6)(m^2+7m+4)}{3m^2-18m+24} = \frac{(m-4)(3m-6)(4m)}{3m-6}$$

$$\frac{(m-4)(3m-6)(2m)}{m-4} - \frac{(m-4)(3m-6)(m^2+7m+4)}{3m^2-18m+24} = \frac{(m-4)(3m-6)(4m)}{3m-6}$$

$$(3m-6)(2m) - (m^2+7m+4) = (m-4)4m$$

$$6m^2 - 12m - m^2 - 7m - 4 = 4m - 16m$$

$$5m^2 - 19m - 4 = 4m - 16m$$

$$m^2 - 3m - 4 = 0$$

$$(m-4)(m+1) = 0$$

$$m-4 = 0 \text{ or } m+1 = 0$$

$$m = 4 \qquad m = -1$$

When solving a rational equation, any possible solution that results in a zero in the denominator must be excluded from the list of solutions.

Check each solution by substituting into the original equation.

Since $m = 4$ results in a zero in the denominator, it is extraneous. So, the solution is -1.

Check

Solve $\frac{3x}{x-4} - \frac{x^2-7x-4}{x^2-16} = \frac{5}{x+4}$.

$x =$?

Learn Types of Studies

In a **survey**, data are collected from responses given by members of a group regarding their characteristics, behaviors, or opinions.

In an **experiment**, the sample is divided into two randomly selected groups, the *experimental group* and the *control group*. The effect on the experimental group is then compared to the control group.

In an **observational study**, members of a sample are measured or observed without being affected by the study.

To determine when to use each type of study, think about how the data will be obtained and how the study will affect the participants. Each method of study relies on random selection of a sample to ensure that it accurately represents the population.

Example 1 Find Probabilities

A student tossed a fair eight-sided die 200 times and recorded the results. Find the theoretical and experimental probabilities of rolling an 8.

Number on Die	Frequency
1	28
2	19
3	24
4	22
5	21
6	18
7	26
8	42

Theoretical Probability

The theoretical probability is what is expected to happen. Because the die is fair, each side has a $\frac{1}{8}$ chance of being the result. Thus, the theoretical probability is $\frac{1}{8}$, or 12.5%.

Experimental Probability

The experimental probability is based on the data collected from the experiment. Because the die was thrown 200 times and 42 of those throws landed on 8, the experimental probability is $\frac{42}{200}$, or 21%.

◆ LESSON 8-2 · Using Statistical Experiments

Mixed Exercises

7. Describe a spinner that could be used to simulate an event with 2 possible outcomes, one with a probability of 60%.

8. How should numbers be assigned to use a random number generator for the following situation?
Probability of choosing blue: 40%
Probability of choosing green: 20%
Probability of choosing red: 30%
Probability of choosing purple: 10%

Example 1 Find a Standard Deviation

TRACK **A coach recorded the times of each track member for a 400-meter race. Find and interpret the standard deviation of the data.**

400m Race Times (seconds)	
57.1	55.9
59.3	54.9
54.6	50.3
55.2	53.5

Step 1 Find the mean, μ.

$$\mu = \frac{57.1 + 59.3 + 54.6 + 55.2 + 55.9 + 54.9 + 50.3 + 53.5}{8}$$
$$= 55.1$$

The mean running time for the team is 55.1 seconds.

Step 2 Find the squares of the differences, $(\mu - x_n)^2$.

$(55.1 - 57.1)^2 = 4.00$ $(55.1 - 55.9)^2 = 0.64$
$(55.1 - 59.3)^2 = 17.64$ $(55.1 - 54.9)^2 = 0.04$
$(55.1 - 54.6)^2 = 0.25$ $(55.1 - 50.3)^2 = 23.04$
$(55.1 - 55.2)^2 = 0.01$ $(55.1 - 53.5)^2 = 2.56$

Step 3 Find the sum.

Find the sum of the values from **Step 2**.

$$4.00 + 17.64 + 0.25 + 0.01 + 0.64 + 0.04 + 23.04 + 2.56$$
$$= 48.18$$

Step 4 Divide by the number of values.

Divide the sum from **Step 3** by the number of running times.

$\frac{48.18}{8} = 6.0225$ This is the variance.

Step 5 Take the square root of the variance.

$\sqrt{6.0225} \approx 2.45$ This is the standard deviation.

The standard deviation is about 2.45. This is small compared with the run times, which means that the majority of the team members' times are close to the mean of 55.1 seconds, and almost all of the times will likely fall within 2 standard deviations of the mean, or between 50.2 and 60.0 seconds.

Check

How do you know that your solution is reasonable? (*Hint:* The majority of the data should be within one standard deviation of the mean, and almost all of the data should be within two.)

Over half of the times are within one standard deviation of the mean, which is the interval [52.65, 57.55] All of the data are within two standard deviations of the mean, or the interval [50.2, 60].

◆ LESSON 8-3 · **Analyzing Population Data**

Example 1

1. BARBER A barber wants to purchase new professional shears from a Web site. The prices of all of the shears are shown in the table. Use the standard deviation formula to find and interpret the standard deviation of the data. Round your answers to the nearest cent.

Cost of Shears ($)			
50	165	55	79
84	68	38	42

2. READING Ms. Sanchez keeps track of the total number of books each student in the book club reads during the school year. Use the standard deviation formula to find and interpret the standard deviation of the data. Round your answers to the nearest tenth.

Books Read		
9	6	12
8	9	14
10	13	8

Example 3 Identify Coterminal Angles

Find an angle with a positive measure and an angle with a negative measure that are coterminal with a 35° angle.

positive angle: $35° + 360° = 395°$

negative angle: $35° - 360° = -325°$

Check

Find an angle with a positive measure and an angle with a negative measure that are coterminal with a −50° angle.

positive angle: ? °

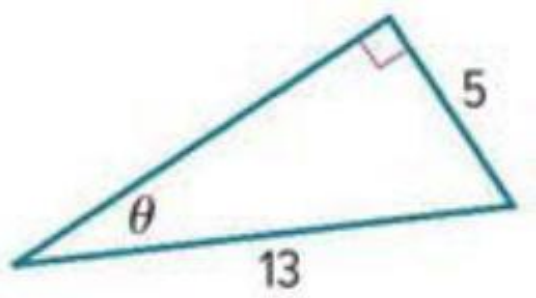
negative angle: ? °

◆ LESSON 9-2 · Trigonometric Functions of General Angles

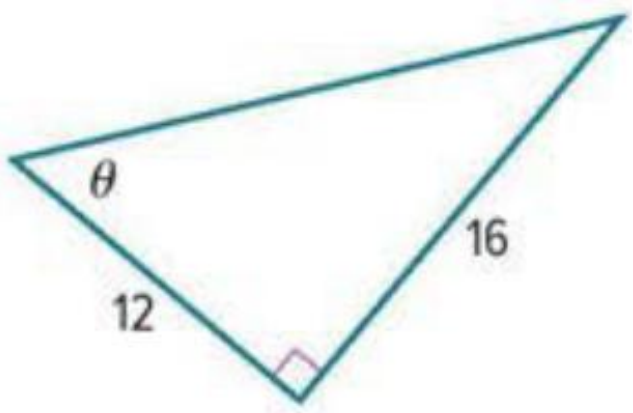
Example 1

Find the exact values of the six trigonometric functions for angle θ.

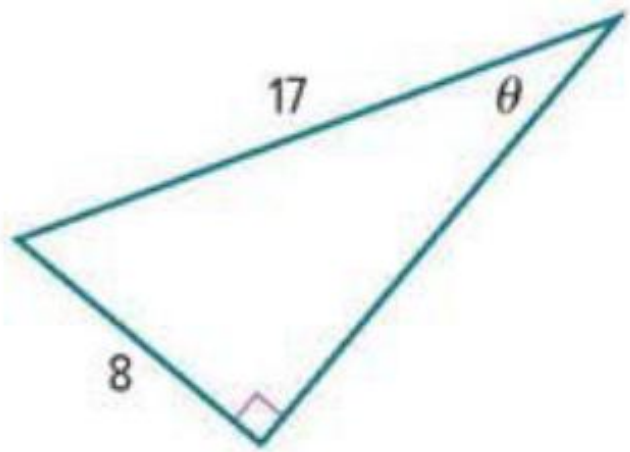
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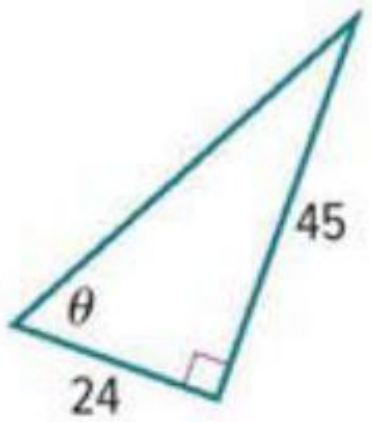


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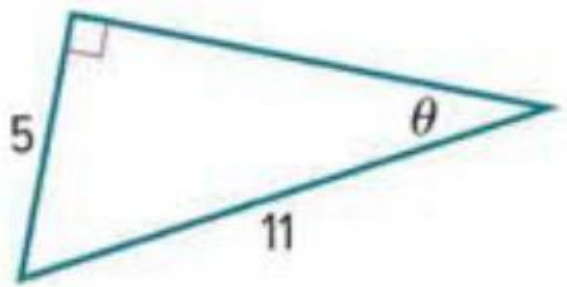


◆ LESSON 9-2 · Trigonometric Functions of General Angles

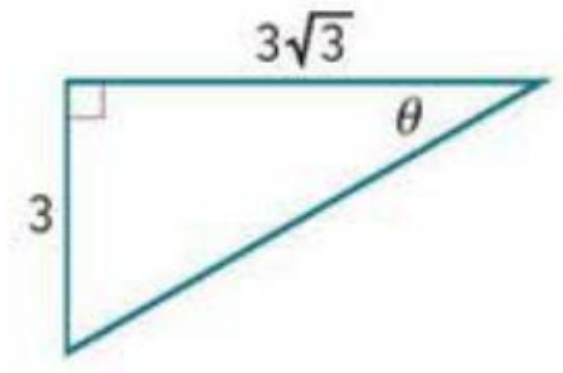
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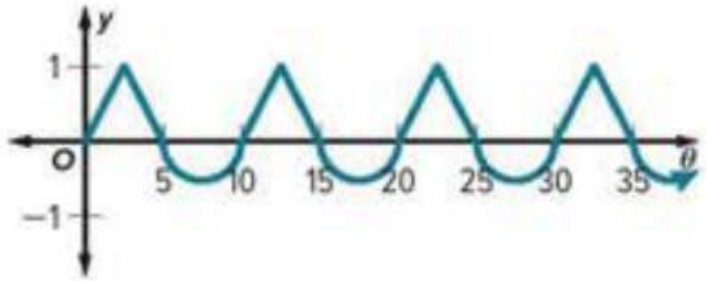
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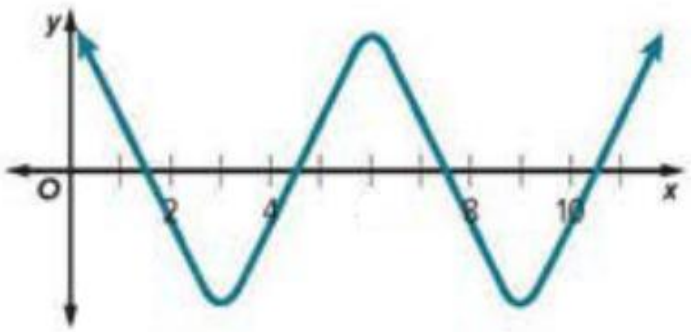
Example 3

Determine the period of the function.

13.

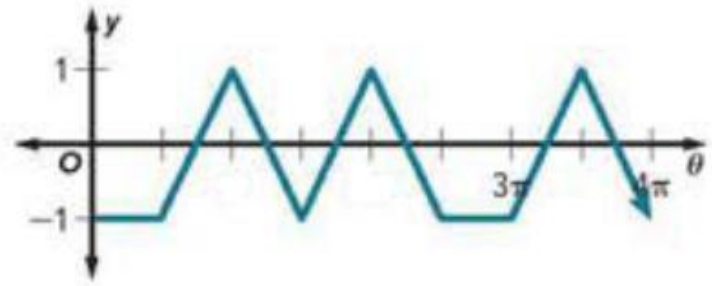


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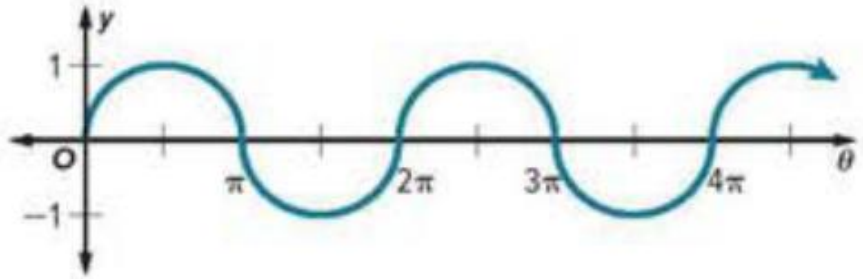


◆ LESSON 9-3 · Circular and Periodic Functions

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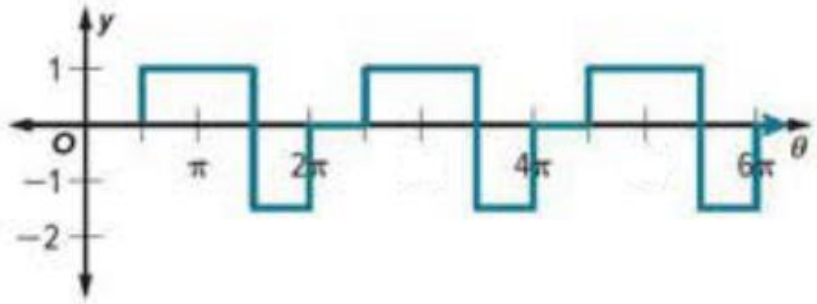
16.



17.



18.



Example 6 Model Periodic Situations

ELECTRICITY The voltage supplied by an electrical outlet can be represented by a periodic function. The voltage oscillates between –120 and 120 volts, with a frequency of 50 cycles per second. Write and graph a function for the voltage v as a function of time t.

Step 1 The voltage oscillates from −120 to 120.

The maximum is 120. The minimum is −120.

The amplitude is 120. $a = 120$

Step 2 The frequency is 50 cycles per second.

The period is $\frac{1}{\text{frequency}}$ or $\frac{1}{50}$.

$$\text{period} = \frac{2\pi}{b} \qquad \text{Period in terms of } b$$

$$\frac{1}{50} = \frac{2\pi}{b} \qquad \text{Substitute.}$$

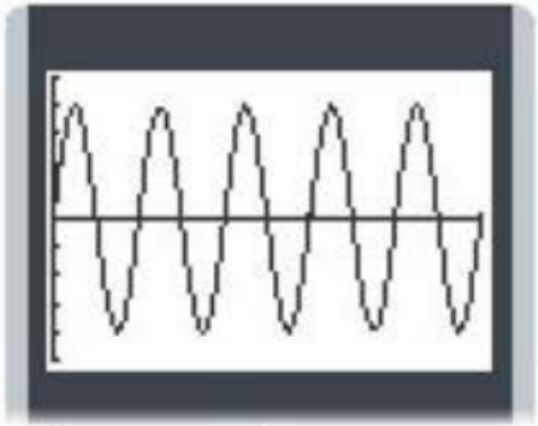
$$b = 100\pi \qquad \text{Solve.}$$

Step 3 Use t instead of x to write the function as a function of time.

$$y = a \sin bt \qquad \text{General equation}$$

$$y = 120 \sin 100\pi t \qquad \text{Substitute.}$$

Step 4 Use a graphing calculator to graph the function.



[0, 0.1] scl: 0.01 by [−150, 150] scl: 30

11. MUL TIPLE CHOICE A spring oscillates with a frequency of 1 cycle per second. The distance between the maximum and minimum points of the oscillation is 3 centimeters. Which of these functions can be used to model the oscillation if y represents the distance in centimeters from the equilibrium position and t is given in seconds? (Lesson 9-4)

A. $y = 1.5 \sin 2\pi t$

B. $y = 1.5 \sin \pi t$

C. $y = 3 \sin 2\pi t$

D. $y = 3 \sin \pi t$

Examples 3 and 4

Find the period and asymptotes of each function, the relative extrema of each secant and cosecant function, and the *x*-intercepts of each cotangent function. Then graph each function.

7. $y = \csc \frac{3}{4}x$

8. $y = \csc 3x$

9. $y = \frac{1}{2} \csc x$

10. $y = \cot 2x$

11. $y = \cot \frac{1}{2}x$

12. $y = -2 \cot x$

◆ LESSON 9-5 · Graphing Other Trigonometric Functions

13. $y = 3 \sec x$

14. $y = \sec \frac{1}{3}x$

15. $y = \sec 4x$

◆ LESSON 9-7 · Inverse Trigonometric Functions

Example 1

Find each value. Write angle measures in degrees and radians.

1. $\text{Cos}^{-1}\left(\frac{\sqrt{3}}{2}\right)$

2. $\text{Sin}^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

3. $\text{Arccos}\left(-\frac{1}{2}\right)$

4. $\text{Arctan}\ \sqrt{3}$

5. $\text{Arccos}\left(-\frac{\sqrt{2}}{2}\right)$

6. $\text{Tan}^{-1}(-1)$

◆ LESSON 9-7 · Inverse Trigonometric Functions

7. $\sin^{-1}\frac{\sqrt{2}}{2}$

8. $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

9. Arcsin 1

Example 2

10. Consider the function $f(x) = 3e^{x-1} + 3$.

a. Graph the function.

b. Determine domain and range.

c. Find the average rate of change over the interval $[-5, -2]$.

11. Consider the function $f(x) = 4e^{2x} - 1$.

a. Graph the function.

b. Determine domain and range.

c. Find the average rate of change over the interval $[-3, -1]$.

12. Consider the function $f(x) = -2e^{x+3} + 2$.

a. Graph the function.

b. Determine domain and range.

c. Find the average rate of change over the interval $[-7, -4]$.

Example 2 Continuous Exponential Decay

SCIENCE The half-life of a radioactive substance is the time it takes for half of the atoms of the substance to disintegrate. The radioactive substance Thorium-230 is used to determine the ages of cave formations and coral. The half-life of Thorium-230 is 75,381 years.



(continued on the next page)

Part A

Determine the value of k and the equation of decay for Thorium-230.

If a is the initial amount of the substance, then the amount y that remains after one half-life period, or 75,381 years, is $\frac{1}{2}a$ or $0.5a$.

$y = ae^{-kt}$	Formula for continuous exponential decay
$0.5a = ae^{-k(75{,}381)}$	$y = 0.5a$ and $t = 75{,}381$
$0.5 = e^{-75{,}381k}$	Divide each side by a.
$\ln 0.5 = \ln e^{-75{,}381k}$	Property of Equality for Logarithmic Equations
$\ln 0.5 = -75{,}381k$	$\ln e^x = x$
$\frac{\ln 0.5}{-75{,}381} = k$	Divide each side by $-75{,}381$.
$0.0000092 \approx k$	Use a calculator.

The rate of decay of Thorium-230 is about 0.0000092, or about 0.00092% per year. Thus, the equation for the decay of Thorium-230 is $y = ae^{-0.0000092t}$.

Part B

How much of a 2-gram sample of Thorium-230 should be left after 1500 years?

$y = ae^{-0.0000092t}$	Formula for the decay of Thorium-230
$= 2e^{-0.0000092(1500)}$	$a = 2$ and $t = 1500$
$= 2e^{-0.0138}$	Simplify.
≈ 1.97	Use a calculator.

After 1500 years, there will be about 1.97 grams of Thorium-230 remaining. Since Thorium-230 has such a long half-life, it is reasonable that after 1500 years only a small amount, 0.03 gram, of the original sample will have decayed.

Check

HALF-LIFE Iodine-131, a radioactive isotope commonly used to treat thyroid cancer, has a half-life of 8.02 days.

Part A

Write the continuous exponential decay equation for Iodine-131. Round the value of k to the nearest thousandth.

Part B

How much of a 15-gram sample will be left after 20 days? Round to the nearest tenth if necessary.

? grams

Example 4 Graph with Oblique Asymptotes

Consider $f(x) = \frac{x^2 + 2x + 1}{x + 4}$.

Part A Find the zeros.

$x^2 + 2x + 1 = 0$ Set $a(x) = 0$.

$(x + 1)^2 = 0$ Factor.

$x + 1 = 0$ Take the square root of each side.

$x = -1$ Subtract 1 from each side.

There is a zero at $x = -1$.

Part B Find the asymptotes.

Find the vertical asymptote.

$x + 4 = 0$ Set $b(x) = 0$.

$x = -4$ Subtract 4 from each side.

There is a vertical asymptote at $x = -4$.

Because the degree of the numerator is greater than the degree of the denominator, there is no horizontal asymptote.

The difference between the degree of the numerator and the degree of the denominator is 1, so there is an oblique asymptote. To find the oblique asymptote, divide the numerator by the denominator.

$$\begin{array}{r} x - 2 \\ x + 4 \overline{)\, x^2 + 2x + 1} \\ (-)\, x^2 + 4x \\ \hline -2x + 1 \\ (-)\, -2x - 8 \\ \hline 9 \end{array}$$

The equation of the asymptote is the quotient excluding the remainder. So, there is an oblique asymptote at $f(x) = x - 2$.

Example 5 Use the Empirical Rule to Analyze Data

A normal distribution has a mean of 42 and a standard deviation of 6.

a. Find the range of values that represents the middle 95% of the distribution.

By the Empirical Rule, the middle 95% of the data in a normal distribution is under the curve between the interval

$[\mu - 2\sigma, \mu + 2\sigma]$.

$\mu - 2\sigma = 42 - 2(6) = 30$ $\mu + 2\sigma = 42 + 2(6) = 54$

b. What percent of the data will be greater than 36?

36 is σ less than μ. Therefore, the range of values greater than 36 is $X > \mu - \sigma$.

This range includes:

- All of the data greater than μ, or 50%.
- The data between $\mu - \sigma$ and μ, or 34%.

The percent of data greater than 36 is 84%.

Example 8

14. TESTING The scores on a test administered to prospective employees are normally distributed with a mean of 100 and a standard deviation of 12.3.

a. What percent of the scores are between 70 and 80?

b. What percent of the scores are over 115?

c. If 75 people take the test, how many would you expect to score lower than 75?

Example 6 Find Arc Length

TRAFFIC A traffic circle, or roundabout, is a circular roadway at the intersection of two or more streets that allows cars to travel through more continuously than a traffic light or stop sign. The diameter of a traffic circle is 160 feet. How far does a car travel in the roundabout if it goes three-fourths of the way around?

Step 1 Find the central angle in radians.

$\theta = \frac{3}{4} \cdot 2\pi$ or $\frac{3\pi}{2}$ The angle is $\frac{3}{4}$ of a rotation.

Step 2 Find the arc length.

$s = r\theta$ Formula for arc length

$= 80 \cdot \frac{3\pi}{2}$ $r = 0.5d = 0.5(160)$ or 80 and $\theta = \frac{3\pi}{2}$

$= 120\pi$ Multiply.

≈ 376.99 Use a calculator.

A car that travels three-fourths of the way around the traffic circle will travel about 377 feet.

Check

SURVEYING If a surveyor's wheel with a radius of 15 inches completes $\frac{13}{20}$ of a rotation, what is the total distance traveled in feet? Round to the nearest hundredth if necessary.

? ft

Example 6

52. TRANSPORTATION A traffic roundabout has a diameter of 200 meters. How far does an automobile travel in the roundabout if it goes one-fourth of the way around?

53. ANALOG CLOCKS The length of the minute hand of an analog clock is 5 inches. If the minute hand rotates from 12 noon to 12:40 P.M., then how far does its point move?

Example 2

State the amplitude, period, and phase shift for each function. Then graph the function and state the domain and range.

7. $y = 3 \cos (\theta - 45°)$

8. $y = 2 \sin (\theta + 60°)$

9. $y = \frac{1}{2} \sin 3\left(\theta - \frac{\pi}{3}\right)$

10. $y = \frac{1}{2} \cos 3\left(\theta - \frac{\pi}{2}\right)$

11. $y = 2 \cos (\theta + 90°)$

12. $y = \frac{1}{4} \sin (\theta - 30°)$